

MULTIMODAL MEDICAL INTELLIGENCE

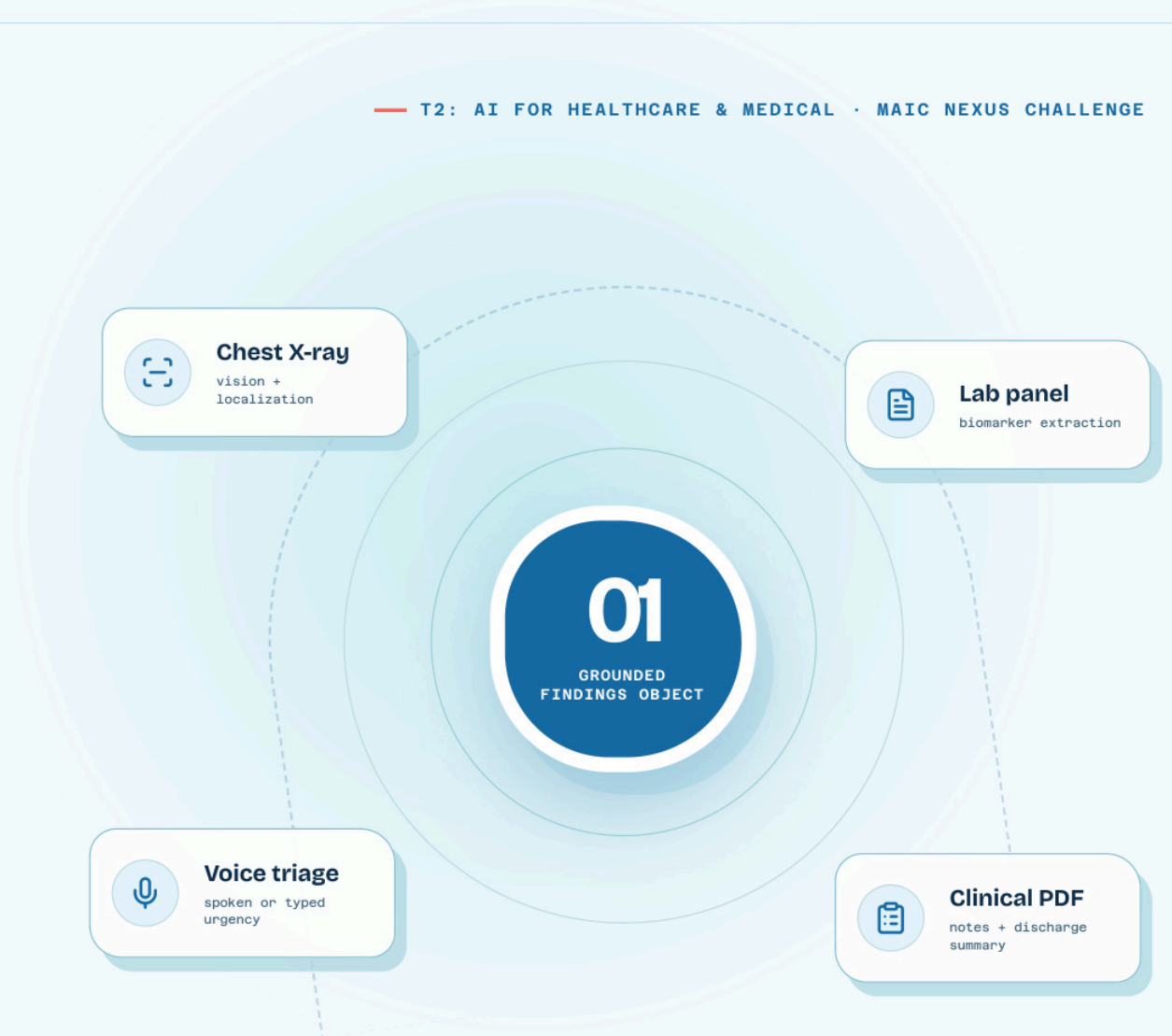
From any medical artifact to actual clinical report.

Three inspectable flows: multimodal analysis with MOH-grounded reports, Ask the scan on this study, and voice triage.

Medical intelligence

Ask the scan

Voice triage



Medical data is multimodal. Every handoff breaks the chain.

Four gaps stand between what clinicians upload and what patients and doctors can safely act on.

● 4 DISCONNECTED MOMENTS

01 / SEE



Imaging stays invisible

Text-first assistants cannot inspect a film or point to the anatomy behind a finding.

HANDOFF

02 / STRUCTURE



Labs stay unstructured

Hb, WBC, and renal panels get summarized as text, not stored as data you can trend.

HANDOFF

03 / GROUND



Context goes missing

Generic answers ignore patient history, Malaysia MOH guidance and local priors.

HANDOFF

04 / EXPLAIN



One voice fits nobody

Clinical language overwhelms patients; simplified output underserves doctors.

THE OPPORTUNITY

Replace four handoffs with one safe, grounded source of truth.

Built for Malaysia

One clinical truth. Two useful voices.

SihatAI routes any artifact into one canonical findings object, then adapts the explanation, not the evidence.

● NO DUPLICATED INFERENCE

 DENSE + INSPECTABLE

Physician view

RLL opacity
Localized · MOH citation [01]

Mild cardiomegaly
Review recommended

Confidence + DDx
Raw scores remain visible

88%



 PLAIN + ACTIONABLE

Patient view

"There is a change in the **lower part of your right lung**. Your doctor should review it together with your symptoms."

Next step · Bring this report and your symptom history to your clinician.

INGEST



ROUTE



ANALYZE



GROUND



GUARD



COMPOSE

The evidence stays in the room.

● CONCEPTUAL PRODUCT VIEW

S

AR

MEDICAL RECORD / MR-2048
Chest X-ray analysis

PA CHEST · 2048 × 2048

ASK THE SCAN

RLL opacity - 91%

Physician Patient

OVERALL CONFIDENCE
88%

ANALYSIS COMPLETE

Right lower lobe opacity **ABNORMAL**
Patchy airspace opacity localized to the right lower zone.

Mild cardiomegaly **REVIEW**
Cardiothoracic ratio is mildly increased.

MOH CPG Community Acquired Pneumonia in Adults **[01]**

↳ Bounding-box localization

↳ Ask the scan

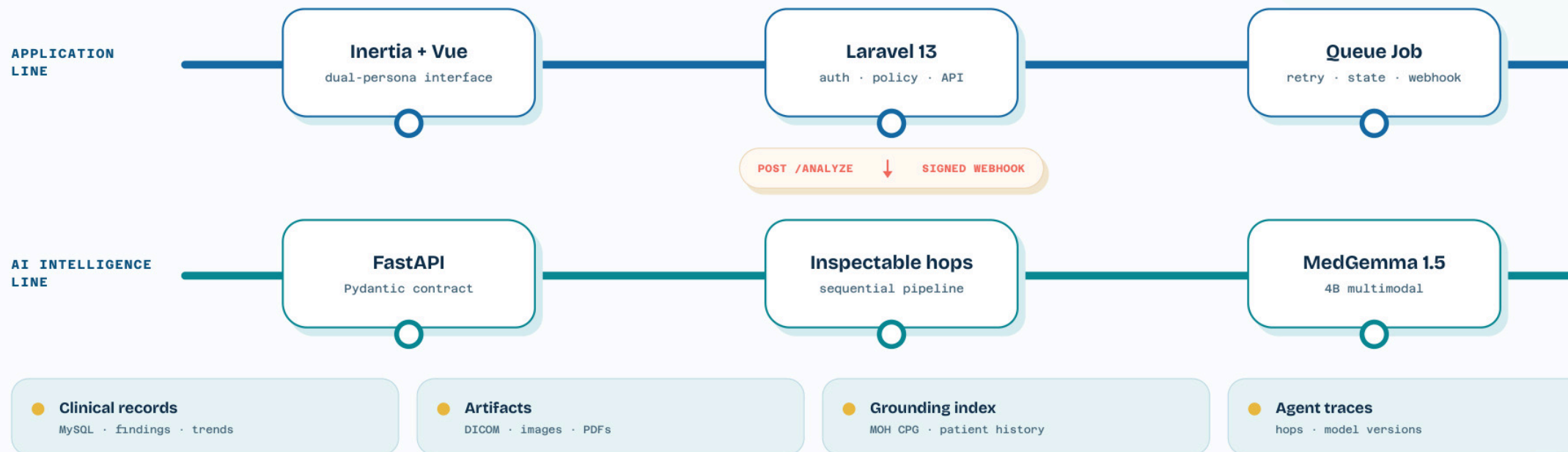
↳ Inline citations

↳ Physician sign-off

Laravel orchestrates. Python reasons.

A clean asynchronous contract lets product engineering and model inference scale independently.

● ONE ASYNC INTERCHANGE



Six specialists. One inspectable relay.

● BOUNDED HOPS · TYPED TOOLS

**De-ID**

scrub PII / PHI

safe_uri

**Router**

detect modality

route_confidence

**Imaging / Doc**

parallel specialists

partial_findings

**Retriever**

BM25 + optional dense

citations[]

**Guardrail**

veto + escalate

ALLOW | WARN

**Composer**

audience adapt

reports()

LIVE TRACE
JOB_7F3A · COMPLETE00
deidentify()
120 ms01
detect_modality()
800 ms02
medgemma_infer()
11.2 s03
rag_retrieve()
900 ms04
safety_check()
450 ms05
compose(locale=bm)
3.1 s

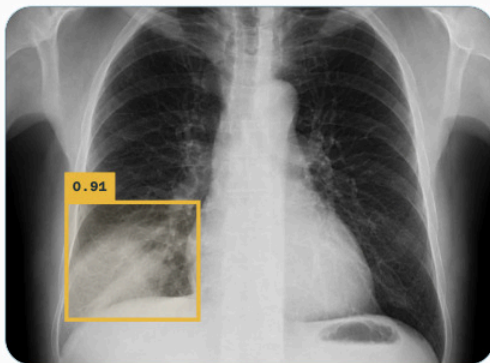
Not a chatbot. A clinical perception stack.

● GOOGLE / MEDGEMMA-1.5-4B-IT

01 · VISION

IMAGE

See the finding

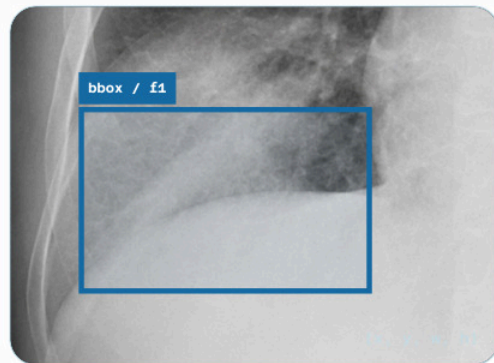


pixel tensor → structured observations[]

02 · GROUNDING

WHERE

Point to anatomy



normalized bbox → viewport-aware overlay

03 · DOCUMENT AI

DATA

Structure the PDF

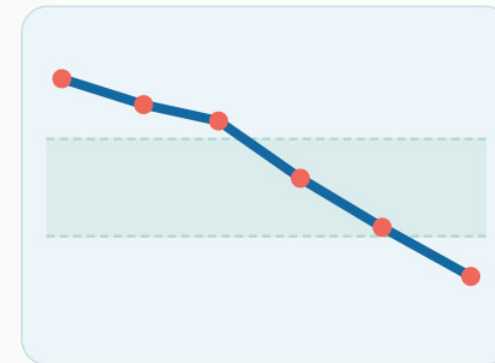
Hb	9.2	LOW
Hct	28.1	LOW
RBC	3.4	LOW
WBC	3.8	LOW
Neut	1.2	LOW
PLT	85	LOW
MCV	72	LOW
CRP	48	HIGH

PDF / OCR → Pydantic schema → MySQL

04 · TEMPORAL

CHANGE

Reason over time



current ⊕ prior → new | stable | resolved

Grounded in evidence. Fluent in Malaysia.

BM

EN

中文

தமிழ்

RETRIEVAL QUERY

"RLL opacity + fever + cough"

01

BM25 search

dense rank when embeddings match

02

MMR rerank

relevance × diversity

03

Confidence gate

abstain below threshold

#1	MOH CPG · Community Acquired Pneumonia	0.89
#2	Patient history · respiratory episode	0.82
#3	MOH CPG · Tuberculosis management	0.76

English

Bahasa Melayu

RINGKASAN UNTUK PESAKIT

Terdapat perubahan pada bahagian bawah paru-paru kanan.

Imej menunjukkan kawasan legap yang mungkin berkaitan dengan jangkitan. Dapatan ini perlu dinilai bersama gejala dan pemeriksaan doktor.

[MOH-CPG-01] Garis Panduan Pneumonia

89% relevance

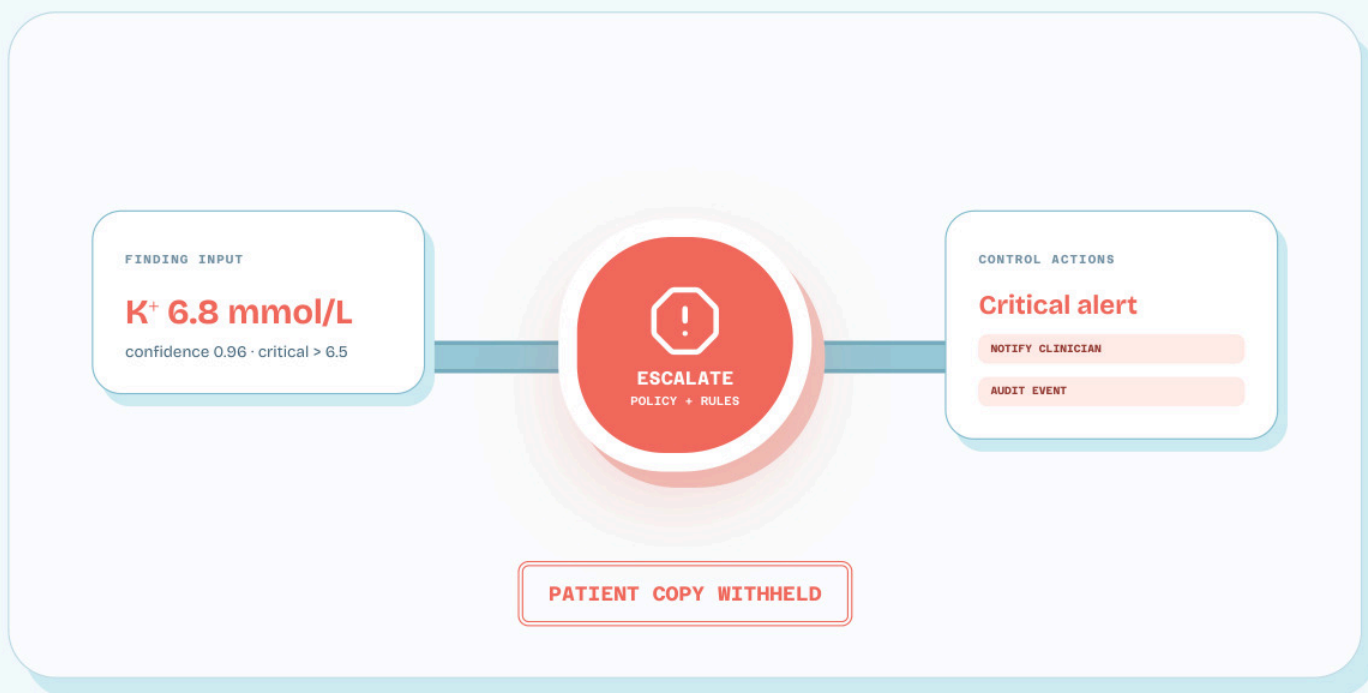
MY-LoRA

QLoRA adapter

Patient Ask the scan phrasing · Bahasa CPG style

Safety is a control plane. Not a disclaimer.

● GUARDRAIL HAS VETO POWER



01

De-identify first

DICOM tags, document PII and metadata are scrubbed before inference.

02

Heuristic confidence

Publish, hedge, or abstain from model-reported scores. Not calibrated probability.

03

Constrain claims

Structured DDx with confidence, never unsupported certainty.

04

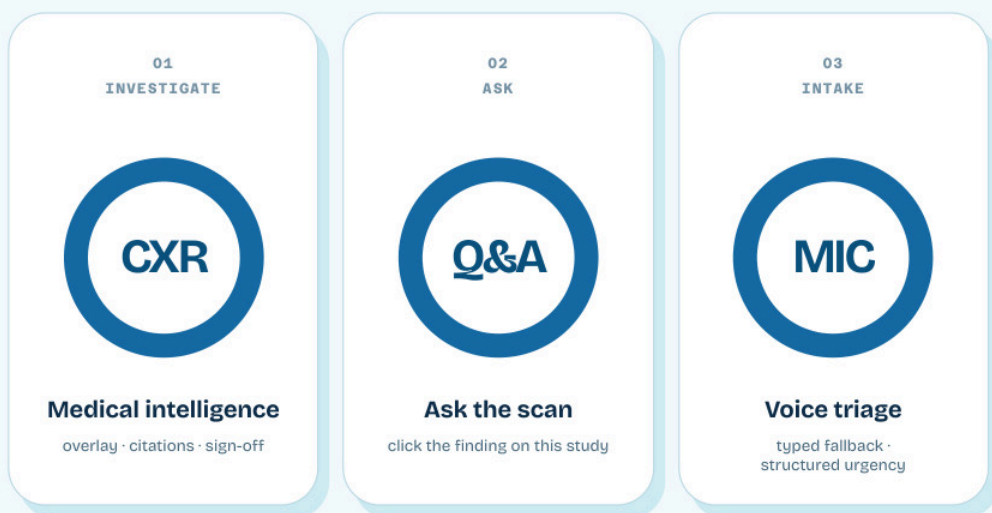
Preserve oversight

Physicians inspect evidence, boxes and citations before sign-off.

Three pillars you can open.

One demo, three inspectable flows.

● SHIPPED PRODUCT



ENGINEERING PROOF

Inspectable workflow



Our advantage is not one model. It is the system around it.

● THREE PILLARS



01

Understands more

X-rays, CT, labs, PDFs and voice all route through one pipeline. Modality detection sends each artifact to the right specialist without re-uploading.

X-RAY · CT · MRI

LAB PDFS

VOICE TRIAGE



02

Grounded in Malaysia

MOH Clinical Practice Guidelines, traceable citations and a localized adapter keep every claim grounded. Reports ship in BM, EN and other local languages.

MOH CPG

CITATIONS

BM · EN · ZH · TA



03

Fails safely

De-ID runs before inference, and confidence bands decide when to publish, hedge or abstain. Guardrails can veto unsafe output and escalate critical values.

DE-ID FIRST

ABSTENTION

VETO + ESCALATE



04

One truth, two audiences

MedGemma runs once to produce a single structured findings object. That same truth becomes a physician report and a patient-friendly summary without duplicated inference.

PHYSICIAN REPORT

PATIENT REPORT

SHARED FINDINGS

SYSTEM OUTCOME CHAIN

Multimodal intake

Any clinical artifact enters one routed pipeline



Inspectable reasoning

Agent hops, traces and citations stay visible



Safer clinical action

Guardrails route uncertainty to clinicians

— THE NEXT INTERFACE IS NOT A CHATBOX

 Understands images

 Escalates safely

 Grounds every claim

 Adapts to people

It is a safe, grounded clinical intelligence layer.

SihatAI combines multimodal analysis, Ask the scan, and voice triage in one inspectable pipeline.

01

Understands

imaging · labs · documents · voice

02

Reasons

hops · RAG · longitudinal context

03

Protects

de-ID · abstention · escalation

04

Adapts

physician · patient · multilingual